MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 01 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If one of the zeroes of the quadratic polynomial $(k-1)x^2 + kx + 1$ is -3, then the value of k is: [1]

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

Q2. A quadratic polynomial, the zeroes of which are -3 and 4, is: [1]

(A) $x^2 - x + 12$ (B) $x^2 + x + 12$ (C) $\frac{x^2}{2} - \frac{x}{2} - 6$ (D) $2x^2 + 2x - 24$

Q3. If the zeroes of the quadratic polynomial $x^2 + (a+1)x + b$ are 2 and -3, then: [1]

(A) $a = -7, b = -1$ (B) $a = 5, b = -1$ (C) $a = 2, b = -6$ (D) $a = 0, b = -6$

Q4. The total number of polynomials having zeroes as -2 and 5 is: [1]

(A) 1 (B) 2 (C) 3 (D) more than 3

Q5. **Assertion (A):** The graph of $p(x) = x^2 - 4x + 4$ touches the x-axis at exactly one point $(2, 0)$.
Reason (R): A quadratic polynomial has two equal zeroes if and only if $D = b^2 - 4ac = 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. Find the zeroes of the quadratic polynomial $f(u) = 4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients. [2]

Q7. Find the zeroes of the quadratic polynomial $f(x) = 6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients. [3]

Q8. Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $\frac{1}{3}$ respectively. [3]

Q9. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 5x + 4$, find the value of: [3]

$$\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$$

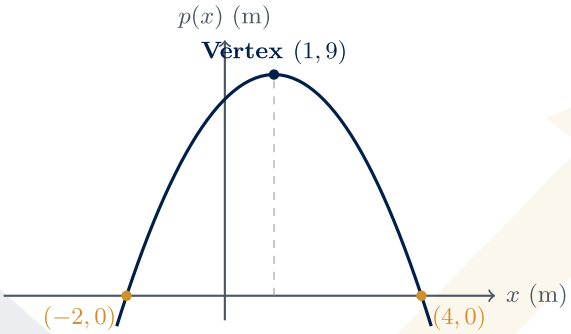
SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — Parabolic Highway Underpass

A highway underpass is designed with a parabolic shape to allow vehicles to pass beneath an overhead railway line. The profile of the arch is represented mathematically by the quadratic polynomial:

$$p(x) = -x^2 + 2x + 8$$

where x represents the horizontal distance (in metres) from the centre line and $p(x)$ is the clearance height (in metres) of the underpass above the road level.



Based on the above mathematical model, answer the following questions:

(i) Find the zeroes of the polynomial $p(x) = -x^2 + 2x + 8$. What do these zeroes represent physically? [1]

(ii) What is the maximum clearance height of the underpass, and at what horizontal position does it occur? [1]

(iii) **(a)** If a second underpass nearby has zeroes at -3 m and 5 m, find the quadratic polynomial representing its shape. [2]

— OR —

(b) A cargo truck of width 2 m (extending from $x = -1$ to $x = 1$) and height 7.5 m enters the underpass along the central lane. Will the truck clear the underpass safely? Justify mathematically. [2]

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